

Martingales — Solutions (set 1)

Problem I — Likelihood Ratio Martingale, Hellinger Affinity, and Change of Reference Law (18 pts)

Throughout the solution we write $W_i = \frac{g(X_i)}{f(X_i)}$, so that the W_i are i.i.d. (measurable functions of the i.i.d. X_i), $W_i \geq 0$, and $L_n = \prod_{i=1}^n W_i$, with $L_n = L_{n-1} W_n$, where L_{n-1} is $\mathcal{F}_{n-1}$ -measurable and W_n is independent of $\mathcal{F}_{n-1}$ (since $X_n \perp (X_1, \dots, X_{n-1})$).

Question 1.

(a) We have $L_n \geq 0$ as a product of nonnegative quantities. Let us compute $\mathbb{E}[W_1]$. Since X_1 has density f with respect to ν , applying the transfer formula with $h = g/f$ (measurable, nonnegative, and everywhere well defined since $f > 0$),

$$\mathbb{E}[W_1] = \mathbb{E}\left[\frac{g(X_1)}{f(X_1)}\right] = \int_{\mathbf{X}} \frac{g(x)}{f(x)} f(x) d\nu(x) = \int_{\mathbf{X}} g d\nu = 1.$$

Since the W_i are independent and *nonnegative*, Tonelli's theorem (factorization of the expectation of a product of independent nonnegative variables) gives, without any prior integrability assumption,

$$\mathbb{E}[L_n] = \mathbb{E}\left[\prod_{i=1}^n W_i\right] = \prod_{i=1}^n \mathbb{E}[W_i] = 1 < \infty.$$

Since $L_n \geq 0$ and $\mathbb{E}[L_n] = 1 < \infty$, we have $L_n \in L^1(\mathbb{P})$ and $\mathbb{E}[L_n] = 1$ for every $n \geq 0$ (the case $n = 0$ being trivial, $L_0 = 1$).

(b) The process (L_n) is adapted (each L_n is a measurable function of $X_1, \dots, X_n$, hence $\mathcal{F}_n$ -measurable) and integrable by (a). For $n \geq 0$, write $L_{n+1} = L_n W_{n+1}$. Since L_n is $\mathcal{F}_n$ -measurable and nonnegative, $W_{n+1} \geq 0$, and $L_{n+1} = L_n W_{n+1} \in L^1$, we may “pull out what is measurable”:

$$\mathbb{E}[L_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[L_n W_{n+1} \mid \mathcal{F}_n] = L_n \mathbb{E}[W_{n+1} \mid \mathcal{F}_n].$$

Now $W_{n+1} = g(X_{n+1})/f(X_{n+1})$ is a function of X_{n+1} , which is independent of $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$. By the independence property of conditional expectation,

$$\mathbb{E}[W_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[W_{n+1}] = 1 \quad \text{a.s.}$$

Hence $\mathbb{E}[L_{n+1} \mid \mathcal{F}_n] = L_n$ a.s., which shows that $(L_n)_{n \geq 0}$ is a martingale.

Question 2.

(a) Let us compute $\mathbb{E}[\sqrt{W_1}]$ by transfer:

$$\mathbb{E}[\sqrt{W_1}] = \int_{\mathbf{X}} \sqrt{\frac{g(x)}{f(x)}} f(x) d\nu(x) = \int_{\mathbf{X}} \sqrt{f(x)g(x)} d\nu(x) = \rho.$$

By the Cauchy–Schwarz inequality in $L^2(\nu)$, applied to $\sqrt{f}$ and $\sqrt{g}$,

$$\rho = \int \sqrt{f} \sqrt{g} d\nu \leq \left(\int f d\nu \right)^{1/2} \left(\int g d\nu \right)^{1/2} = 1,$$

and $\rho \geq 0$ as the integral of a nonnegative function; hence $0 \leq \rho \leq 1$. In particular $\mathbb{E}[\sqrt{W_1}] = \rho \leq 1 < \infty$. Next $\sqrt{L_n} = \prod_{i=1}^n \sqrt{W_i}$ is a product of independent nonnegative variables, whence, by Tonelli as in 1(a):

$$\mathbb{E}[\sqrt{L_n}] = \prod_{i=1}^n \mathbb{E}[\sqrt{W_i}] = \rho^n.$$

(b) The function $\psi(t) = \sqrt{t}$ is *concave* on $[0, \infty[$. The process $(\sqrt{L_n})$ is adapted and integrable (since $\mathbb{E}[\sqrt{L_n}] = \rho^n \leq 1 < \infty$). By the conditional Jensen inequality applied to the concave function ψ (with the reversed direction),

$$\mathbb{E}[\sqrt{L_{n+1}} \mid \mathcal{F}_n] = \mathbb{E}[\psi(L_{n+1}) \mid \mathcal{F}_n] \leq \psi(\mathbb{E}[L_{n+1} \mid \mathcal{F}_n]) = \sqrt{L_n} \quad \text{a.s.},$$

the last equality following from the martingale property (Q1b). Hence $(\sqrt{L_n})_{n \geq 0}$ is a supermartingale.

Equivalent direct computation. With $\sqrt{L_{n+1}} = \sqrt{L_n} \sqrt{W_{n+1}}$, $\sqrt{L_n}$ being $\mathcal{F}_n$ -measurable and nonnegative and $\sqrt{W_{n+1}}$ independent of $\mathcal{F}_n$,

$$\mathbb{E}[\sqrt{L_{n+1}} \mid \mathcal{F}_n] = \sqrt{L_n} \mathbb{E}[\sqrt{W_{n+1}}] = \rho \sqrt{L_n} \leq \sqrt{L_n},$$

which recovers the supermartingale property (and makes the “slope” ρ explicit).

(c) In the Cauchy–Schwarz inequality above, equality $\rho = 1$ holds if and only if $\sqrt{f}$ and $\sqrt{g}$ are proportional ν -a.e., say $\sqrt{g} = c\sqrt{f}$ ν -a.e. with $c \geq 0$. Integrating $g = c^2 f$, we obtain $1 = \int g d\nu = c^2 \int f d\nu = c^2$, so $c = 1$, i.e. $f = g$ ν -a.e. The assumption $\nu(\{f \neq g\}) > 0$ rules out this case, so the inequality is strict: $\rho < 1$.

The process (L_n) is a nonnegative martingale, hence in particular a nonnegative supermartingale; by the convergence theorem for nonnegative supermartingales (admitted), there exists a random variable $L_\infty \geq 0$ such that $L_n \rightarrow L_\infty$ a.s. By continuity, $\sqrt{L_n} \rightarrow \sqrt{L_\infty}$ a.s., and Fatou’s lemma gives

$$\mathbb{E}[\sqrt{L_\infty}] \leq \liminf_n \mathbb{E}[\sqrt{L_n}] = \liminf_n \rho^n = 0,$$

since $0 \leq \rho < 1$. Hence $\mathbb{E}[\sqrt{L_\infty}] = 0$ with $\sqrt{L_\infty} \geq 0$, which forces $\sqrt{L_\infty} = 0$ a.s., whence $L_\infty = 0$ a.s. Thus $L_n \rightarrow 0$ almost surely.

Question 3.

(a) Under $\mathbb{Q}$, the X_i are i.i.d. with density g . Set $V_i = f(X_i)/g(X_i)$ (everywhere well defined since $g > 0$), so that $M_n = \prod_{i=1}^n V_i$ and the V_i are, under $\mathbb{Q}$, i.i.d. and nonnegative. By transfer under $\mathbb{Q}$ (density g),

$$\mathbb{E}_{\mathbb{Q}}[V_1] = \int_{\mathbf{X}} \frac{f(x)}{g(x)} g(x) d\nu(x) = \int_{\mathbf{X}} f d\nu = 1.$$

The process (M_n) is adapted, nonnegative, and by Tonelli $\mathbb{E}_{\mathbb{Q}}[M_n] = \prod_i \mathbb{E}_{\mathbb{Q}}[V_i] = 1 < \infty$, hence $\mathbb{Q}$ -integrable. Since $M_{n+1} = M_n V_{n+1}$ with M_n $\mathcal{F}_n$ -measurable and nonnegative and V_{n+1} independent of $\mathcal{F}_n$ under $\mathbb{Q}$,

$$\mathbb{E}_{\mathbb{Q}}[M_{n+1} \mid \mathcal{F}_n] = M_n \mathbb{E}_{\mathbb{Q}}[V_{n+1}] = M_n \quad \mathbb{Q}\text{-a.s.}$$

Hence (M_n) is a martingale under $\mathbb{Q}$. (This is exactly the likelihood ratio martingale of Q1, but with the roles of f and g swapped.)

(b) Under $\mathbb{Q}$, let us compute $\mathbb{E}_{\mathbb{Q}}[W_1]$ where $W_1 = g(X_1)/f(X_1)$, by transfer with density g :

$$\mathbb{E}_{\mathbb{Q}}[W_1] = \int_{\mathbf{X}} \frac{g(x)}{f(x)} g(x) d\nu(x) = \int_{\mathbf{X}} \frac{g(x)^2}{f(x)} d\nu(x) = \chi.$$

Set $\phi(x) = g(x)/f(x)$. By Cauchy–Schwarz (equivalently, Jensen’s inequality for the law $g d\nu$ and the convex function $t \mapsto t^2$),

$$1 = \left(\int g d\nu \right)^2 = \left(\int \phi g d\nu \right)^2 \leq \left(\int \phi^2 g d\nu \right) \left(\int 1 g d\nu \right) = \chi \cdot 1 = \chi,$$

so $\chi \geq 1$. Since $L_{n+1} = L_n W_{n+1}$, with L_n being $\mathcal{F}_n$ -measurable and nonnegative and W_{n+1} , under $\mathbb{Q}$, independent of $\mathcal{F}_n$ and $\mathbb{Q}$ -integrable ($\mathbb{E}_{\mathbb{Q}}[W_{n+1}] = \chi < \infty$),

$$\mathbb{E}_{\mathbb{Q}}[L_{n+1} \mid \mathcal{F}_n] = L_n \mathbb{E}_{\mathbb{Q}}[W_{n+1}] = \chi L_n \quad \mathbb{Q}\text{-a.s.}$$

Since $\chi \geq 1$ and $L_n \geq 0$, we have $\mathbb{E}_{\mathbb{Q}}[L_{n+1} \mid \mathcal{F}_n] = \chi L_n \geq L_n$; moreover (L_n) is $\mathbb{Q}$ -integrable since $\mathbb{E}_{\mathbb{Q}}[L_n] = \chi^n < \infty$. Hence (L_n) is a submartingale under $\mathbb{Q}$.

Question 4.

Under $\mathbb{P}$, the process (L_n) is a martingale (Q1b), φ is convex with $\varphi(L_n) \in L^1(\mathbb{P})$ by assumption, and $(\varphi(L_n))$ is adapted. By the conditional Jensen inequality for the convex function φ ,

$$\mathbb{E}[\varphi(L_{n+1}) \mid \mathcal{F}_n] \geq \varphi(\mathbb{E}[L_{n+1} \mid \mathcal{F}_n]) = \varphi(L_n) \quad \text{a.s.},$$

using $\mathbb{E}[L_{n+1} \mid \mathcal{F}_n] = L_n$. Hence $(\varphi(L_n))_{n \geq 0}$ is a submartingale.

Taking expectations and by the tower property,

$$\mathbb{E}[\varphi(L_{n+1})] = \mathbb{E}[\mathbb{E}[\varphi(L_{n+1}) \mid \mathcal{F}_n]] \geq \mathbb{E}[\varphi(L_n)],$$

so the sequence $n \mapsto \mathbb{E}[\varphi(L_n)]$ is nondecreasing.

Application. The function $\varphi(t) = t \log^+ t$ (with the convention $0 \log^+ 0 = 0$, and $\log^+ t = \max(\log t, 0)$) is convex on $[0, \infty[$. Indeed, $\varphi(t) = 0$ on $[0, 1]$ and $\varphi(t) = t \log t$ on $[1, \infty[$, which is convex ($\varphi''(t) = 1/t > 0$); the two pieces join continuously at $t = 1$, and the right derivative $\varphi'(1^+) = \log 1 + 1 = 1$ exceeds the left derivative $\varphi'(1^-) = 0$: the slope is globally nondecreasing, so φ is convex on $[0, \infty[$. Under the integrability assumption $\mathbb{E}[L_n \log^+ L_n] < \infty$ for every n , the general result applies and $n \mapsto \mathbb{E}[L_n \log^+ L_n]$ is *nondecreasing*.

Remark. Denote by $\mathbb{P}_n, \mathbb{Q}_n$ the laws of $(X_1, \dots, X_n)$ under $\mathbb{P}$ and $\mathbb{Q}$. Then $L_n = d\mathbb{Q}_n/d\mathbb{P}_n$, and $\mathbb{E}_{\mathbb{P}}[L_n \log L_n] = \text{KL}(\mathbb{Q}_n \parallel \mathbb{P}_n)$ is the Kullback–Leibler divergence accumulated over the first n observations: the monotonicity above reflects the fact that the information distinguishing $\mathbb{Q}$ from $\mathbb{P}$ can only increase with the number of observations. (Here we worked with $\log^+$ rather than $\log$ to guarantee convexity on all of $[0, \infty[$; the difference $\mathbb{E}[L_n(\log^+ L_n - \log L_n)] = \mathbb{E}[L_n \log^- L_n]$ is bounded, which does not affect the interpretation.)

Problem II — Ruin Probability of a Skip-Free-Downward Random Walk with Positive Drift (20 pts)

Preliminary. Note that $|X_1| \leq 2$, so X_1 is integrable and $\mu = \mathbb{E}[X_1] = 2p - q = 3p - 1$. The assumption $p > \frac{1}{3}$ gives $\mu > 0$.

Question 1.

For every $n \in \mathbb{N}$, the variable $S_n = X_1 + \cdots + X_n$ is $\mathcal{F}_n$ -measurable, hence (S_n) is adapted. For every $n \in \mathbb{N}$,

$$\{\tau_{-a} = n\} = \{S_0 \neq -a\} \cap \cdots \cap \{S_{n-1} \neq -a\} \cap \{S_n = -a\} \in \mathcal{F}_n,$$

since each event $\{S_k = -a\}$ (resp. $\{S_k \neq -a\}$) belongs to $\mathcal{F}_k \subset \mathcal{F}_n$. Thus $\{\tau_{-a} \leq n\} = \bigcup_{k=0}^n \{\tau_{-a} = k\} \in \mathcal{F}_n$, which shows that τ_{-a} is a stopping time.

Exactness of the hitting. The increments equal $+2$ or -1 , so $S_{n+1} - S_n \in \{+2, -1\}$. Starting from $S_0 = 0 \geq -a + 1$ (since $a \geq 1$), as long as $S_n > -a$ we have $S_n \geq -a + 1$ (the values of S are integers), and the only possible downward jump is -1 ; at such a jump $S_{n+1} = S_n - 1 \geq -a$. The walk therefore cannot pass from $\{S_n > -a\}$ to $\{S_{n+1} < -a\}$: the first value $\leq -a$ that is reached is necessarily $-a$. Hence $S_{\tau_{-a}} = -a$ on $\{\tau_{-a} < \infty\}$. (This is the “skip-free downward” property.)

Question 2.

(a) The process $M_n = e^{\theta S_n}$ is adapted, and since $-n \leq S_n \leq 2n$, we have $0 < M_n \leq e^{2|\theta|n}$: M_n is bounded, hence integrable for each n . Since X_{n+1} is independent of $\mathcal{F}_n$ and S_n is $\mathcal{F}_n$ -measurable,

$$\mathbb{E}[M_{n+1} | \mathcal{F}_n] = e^{\theta S_n} \mathbb{E}[e^{\theta X_{n+1}} | \mathcal{F}_n] = M_n \mathbb{E}[e^{\theta X_1}] = M_n(p e^{2\theta} + q e^{-\theta}).$$

Hence (M_n) is a martingale if and only if $p e^{2\theta} + q e^{-\theta} = 1$. Setting $y = e^\theta > 0$ and multiplying by y :

$$p y^3 + q - y = 0, \quad \text{that is} \quad p y^3 - y + q = 0.$$

(b) Set $P(y) = p y^3 - y + q$. We have $P(1) = p - 1 + q = 0$, so $y = 1$ is a root and we factor:

$$P(y) = (y - 1)(p y^2 + p y - q).$$

Verification: $(y - 1)(p y^2 + p y - q) = p y^3 + p y^2 - q y - p y^2 - p y + q = p y^3 - (p + q)y + q = p y^3 - y + q$, since $p + q = 1$.

The other roots satisfy $p y^2 + p y - q = 0$, hence

$$y = \frac{-p \pm \sqrt{p^2 + 4pq}}{2p}.$$

The root with the $-$ sign is strictly negative (numerator < 0), hence not admissible (since $y = e^\theta > 0$). The root with the $+$ sign,

$$y_* = \frac{-p + \sqrt{p^2 + 4pq}}{2p},$$

is strictly positive. Let us show $y_* < 1$: set $Q(y) := p y^2 + p y - q$ (a convex parabola). We have $Q(0) = -q < 0$ and $Q(1) = p + p - q = 2p - q = \mu > 0$ (since $p > \frac{1}{3}$). By the intermediate value theorem Q vanishes at a point of $]0, 1[$, and this is its only positive root (the product of the two roots of Q equals $-q/p < 0$, so one of them is negative). This positive root is exactly y_* , and it lies in $]0, 1[$. Since the third root of P is 1 and the other root of Q is negative, y_* is the unique root of P in $]0, 1[$. We set $\theta_* = \ln y_* < 0$ and $M_n = y_*^{S_n}$.

Question 3.

(a) Since τ_{-a} is a stopping time and (M_n) is a martingale, the stopped process $(M_{n \wedge \tau_{-a}})_n$ is again a martingale (stability of the martingale property under stopping at a stopping time). *Boundedness:* by Question 1, for every $m \leq \tau_{-a}$ we have $S_m \geq -a$; hence for every n , $S_{n \wedge \tau_{-a}} \geq -a$. Since $0 < y_* < 1$, the map $u \mapsto y_*^u$ is strictly decreasing on $\mathbb{R}$, whence

$$0 < M_{n \wedge \tau_{-a}} = y_*^{S_{n \wedge \tau_{-a}}} \leq y_*^{-a} =: C < \infty.$$

The stopped martingale is therefore bounded by the constant $C = y_*^{-a}$.

(b) On $\{\tau_{-a} < \infty\}$: for $n \geq \tau_{-a}$ we have $n \wedge \tau_{-a} = \tau_{-a}$, so $M_{n \wedge \tau_{-a}} = y_*^{S_{\tau_{-a}}} = y_*^{-a}$ (constant in n once $n \geq \tau_{-a}$, using $S_{\tau_{-a}} = -a$ from Question 1); the limit therefore equals y_*^{-a} .

On $\{\tau_{-a} = \infty\}$: then $n \wedge \tau_{-a} = n$ and $M_{n \wedge \tau_{-a}} = y_*^{S_n}$. By the law of large numbers $S_n/n \rightarrow \mu > 0$ a.s., so $S_n \rightarrow +\infty$ a.s. (in particular on $\{\tau_{-a} = \infty\}$, as granted). Since $0 < y_* < 1$, $y_*^{S_n} \rightarrow 0$.

In both cases, a.s.,

$$M_{n \wedge \tau_{-a}} \xrightarrow[n \rightarrow \infty]{} y_*^{-a} \mathbf{1}_{\{\tau_{-a} < \infty\}}$$

(the indicator being 0 on $\{\tau_{-a} = \infty\}$, the expression is well defined there).

(c) The sequence $(M_{n \wedge \tau_{-a}})_n$ is dominated by the integrable constant $C = y_*^{-a}$; by the dominated convergence theorem, its a.s. limit is also a limit in L^1 , and in particular

$$\mathbb{E} \left[\lim_n M_{n \wedge \tau_{-a}} \right] = \lim_n \mathbb{E}[M_{n \wedge \tau_{-a}}].$$

By the martingale property (and $M_0 = y_*^{S_0} = y_*^0 = 1$), $\mathbb{E}[M_{n \wedge \tau_{-a}}] = \mathbb{E}[M_0] = 1$ for all n . On the other hand, by (b),

$$\mathbb{E} \left[\lim_n M_{n \wedge \tau_{-a}} \right] = \mathbb{E} [y_*^{-a} \mathbf{1}_{\{\tau_{-a} < \infty\}}] = y_*^{-a} \rho.$$

We obtain $y_*^{-a} \rho = 1$, that is,

$$\rho = \mathbb{P}(\tau_{-a} < \infty) = y_*^a.$$

(Rigour of the passage to the limit: we do not apply the optional stopping theorem to τ_{-a} directly — which is unbounded and may be infinite — but at the deterministic times n via the bounded stopped martingale, passing to the limit by dominated convergence. This is the key point.)

Question 4.

(a) *Hitting with a lower-bounded probability in a fixed number of steps.* Choose $k = a + b$. On $\{\tau > n\}$ we have $-a < S_n < b$, so $S_n \geq -a + 1$. If the next k increments all equal $+2$, then

$$S_{n+k} = S_n + 2k \geq (-a + 1) + 2(a + b) = a + 2b + 1 \geq b,$$

so $\tau \leq n + k$. Now, by independence of the X_i ,

$$\mathbb{P}(X_{n+1} = \dots = X_{n+k} = +2 \mid \mathcal{F}_n) = p^k =: q_0 \in]0, 1[.$$

Hence on $\{\tau > n\}$, $\mathbb{P}(\tau \leq n + k \mid \mathcal{F}_n) \geq q_0$ a.s., i.e. $\mathbb{P}(\tau > n + k \mid \mathcal{F}_n) \leq 1 - q_0$. In particular, for every $m \geq 0$,

$$\mathbb{P}(\tau > (m+1)k) = \mathbb{E}[\mathbf{1}_{\{\tau > (m+1)k\}} \mathbb{P}(\tau > (m+1)k \mid \mathcal{F}_{mk})] \leq (1 - q_0) \mathbb{P}(\tau > mk),$$

whence by induction $\mathbb{P}(\tau > mk) \leq (1 - q_0)^m$. The tail of τ decreases geometrically, so

$$\mathbb{E}[\tau] = \sum_{n \geq 0} \mathbb{P}(\tau > n) \leq k \sum_{m \geq 0} \mathbb{P}(\tau > mk) \leq k \sum_{m \geq 0} (1 - q_0)^m = \frac{k}{q_0} < \infty.$$

(We used that $n \mapsto \mathbb{P}(\tau > n)$ is decreasing in order to group the terms into blocks of length k .)

(b) *Application of Wald.* The X_n are i.i.d., integrable ($|X_1| \leq 2$), and τ is a stopping time with $\mathbb{E}[\tau] < \infty$: these are exactly the hypotheses of Wald's identity (first moment). It guarantees that S_τ is integrable and that

$$\mathbb{E}[S_\tau] = \mu \mathbb{E}[\tau] = (3p - 1) \mathbb{E}[\tau].$$

Since $\mu = 3p - 1 > 0$, we deduce

$$\boxed{\mathbb{E}[\tau] = \frac{\mathbb{E}[S_\tau]}{3p - 1}}$$

Remark: here S_τ equals $-a$ (exact hitting at the bottom) or a value in $\{b, b + 1\}$ at the top, since a last jump of $+2$ may overshoot b by one unit; the right-hand side is therefore not explicit without information on this overshoot, but the identity $\mathbb{E}[\tau] = \mathbb{E}[S_\tau]/\mu$ is rigorously justified.

Question 5.

(a) For $p = \frac{2}{5}$, $q = \frac{3}{5}$. The equation $py^2 + py - q = 0$ reads $\frac{2}{5}y^2 + \frac{2}{5}y - \frac{3}{5} = 0$, that is, multiplying by $\frac{5}{2}$,

$$y^2 + y - \frac{3}{2} = 0 \iff 2y^2 + 2y - 3 = 0.$$

The positive root is

$$y_* = \frac{-2 + \sqrt{4 + 24}}{4} = \frac{-2 + \sqrt{28}}{4} = \frac{-1 + \sqrt{7}}{2} \approx 0.8229 \in]0, 1[.$$

For $a = 3$:

$$\rho = \mathbb{P}(\tau_{-3} < \infty) = y_*^3 = \left(\frac{-1 + \sqrt{7}}{2} \right)^3 \approx 0.557.$$

Closed form: from $y_*^2 = \frac{3}{2} - y_*$ we get $y_*^3 = y_* y_*^2 = \frac{3}{2} y_* - y_*^2 = \frac{3}{2} y_* - (\frac{3}{2} - y_*) = \frac{5}{2} y_* - \frac{3}{2} = \frac{5}{2} \cdot \frac{-1 + \sqrt{7}}{2} - \frac{3}{2} = \frac{5\sqrt{7} - 11}{4} \approx 0.557$.

(b) Yes: despite a strictly positive drift $\mu = \frac{1}{5} > 0$, we have $\rho = y_*^3 \approx 0.557 < 1$. The walk thus reaches the deficit -3 with probability ≈ 0.557 , and runs off to $+\infty$ without ever reaching it with probability ≈ 0.443 .

Role of the asymmetry. The positive drift forces $S_n \rightarrow +\infty$ a.s., but this does not prevent the walk from first going fairly low: with positive probability it accumulates -1 jumps before climbing back up. The crucial point is that the descents occur *exactly* by steps of -1 : the walk cannot “miss” the level $-a$ from below, so ruin corresponds to an exact hitting ($S_{\tau_{-a}} = -a$), which makes the martingale $y_*^{S_n}$ bounded along the stopped trajectory and allows the exact computation $\rho = y_*^a$. Since $y_* \in]0, 1[$ as soon as $p > \frac{1}{3}$, we always have $\rho = y_*^a < 1$ for $a \geq 1$, and $\rho \rightarrow 0$ as $a \rightarrow \infty$: the deeper the target deficit, the less likely it is to be reached. The boundary $p = \frac{1}{3}$ (zero drift) corresponds to $y_* = 1$, hence $\rho = 1$; and for a negative drift $p < \frac{1}{3}$ we no longer have $S_n \rightarrow +\infty$ and the walk reaches $-a$ almost surely, i.e. $\rho = 1$ again. Thus the strictly positive drift is exactly what allows $\rho < 1$: it is the only regime in which the gambler can “lose” (the walk escapes to $+\infty$).

Problem III — Random Products of Rademacher Factors: Kakutani's Dichotomy (21 pts)

Preliminary. For all k , since $a_k \in [0, 1)$ and $\xi_k \in \{-1, 1\}$, we have $1 + a_k \xi_k \in \{1 - a_k, 1 + a_k\} \subset (0, 2)$, so $M_n > 0$ and M_n is bounded (by $\prod_{k=1}^n (1 + a_k) \leq 2^n$), hence integrable. Moreover M_n is $\mathcal{F}_n$ -measurable.

1. Positive martingale and almost sure convergence.

Adaptedness and integrability: shown above ($M_n > 0$, $\mathcal{F}_n$ -measurable, bounded hence in L^1).

Martingale property: for $n \geq 1$, $M_n = M_{n-1}(1 + a_n \xi_n)$, where M_{n-1} is $\mathcal{F}_{n-1}$ -measurable and bounded, and ξ_n is independent of $\mathcal{F}_{n-1}$. By taking out what is measurable and then using independence,

$$\mathbb{E}[M_n | \mathcal{F}_{n-1}] = M_{n-1} (1 + a_n \mathbb{E}[\xi_n | \mathcal{F}_{n-1}]) = M_{n-1} (1 + a_n \mathbb{E}[\xi_n]) = M_{n-1},$$

since $\mathbb{E}[\xi_n] = \frac{1}{2} - \frac{1}{2} = 0$. Hence (M_n) is a martingale. Taking expectations, $\mathbb{E}[M_n] = \mathbb{E}[M_0] = 1$ for all n .

Almost sure convergence: (M_n) is a positive martingale, hence a positive supermartingale; moreover it is bounded in L^1 since $\sup_n \mathbb{E}[M_n] = \mathbb{E}[M_0] = 1 < \infty$. By the almost sure convergence theorem for (super)martingales (Theorem 5.1, a consequence of Doob's upcrossing lemma), M_n converges almost surely to a random variable M_∞ . Since $M_n \geq 0$, we have $M_\infty \geq 0$ a.s., and by Fatou's lemma

$$\mathbb{E}[M_\infty] = \mathbb{E}[\liminf_n M_n] \leq \liminf_n \mathbb{E}[M_n] = 1 < \infty,$$

so M_∞ is integrable. □

2. Second moment and L^2 convergence.

Since $M_n^2 = \prod_{k=1}^n (1 + a_k \xi_k)^2$ and the factors $(1 + a_k \xi_k)^2$ are independent (functions of the independent ξ_k), with $\mathbb{E}[\xi_k] = 0$, $\mathbb{E}[\xi_k^2] = 1$:

$$\mathbb{E}[M_n^2] = \prod_{k=1}^n \mathbb{E}[(1 + a_k \xi_k)^2] = \prod_{k=1}^n (1 + 2a_k \mathbb{E}[\xi_k] + a_k^2 \mathbb{E}[\xi_k^2]) = \prod_{k=1}^n (1 + a_k^2).$$

(a) Assume $\sum_{k \geq 1} a_k^2 < \infty$. Since $1 + x \leq e^x$ for $x \geq 0$,

$$\sup_n \mathbb{E}[M_n^2] = \prod_{k \geq 1} (1 + a_k^2) \leq \exp\left(\sum_{k \geq 1} a_k^2\right) < \infty,$$

so (M_n) is bounded in L^2 . By the L^p convergence theorem for martingales with $p = 2$ (Theorem 5.3): a martingale bounded in L^p , $p > 1$, converges a.s. and in L^p to a limit $M_\infty \in L^p$. Hence $M_n \rightarrow M_\infty$ in L^2 , and the L^2 limit coincides with the a.s. limit obtained in 1. The L^2 convergence implies L^1 convergence (Cauchy-Schwarz: $\|M_n - M_\infty\|_1 \leq \|M_n - M_\infty\|_2$).

(b) Since $M_n \rightarrow M_\infty$ in L^1 , $\mathbb{E}[M_\infty] = \lim_n \mathbb{E}[M_n] = 1$. Since $M_n \rightarrow M_\infty$ in L^2 , $\mathbb{E}[M_\infty^2] = \lim_n \mathbb{E}[M_n^2] = \prod_{k \geq 1} (1 + a_k^2)$. (In particular $\mathbb{E}[M_\infty] = 1 > 0$, so M_∞ is not a.s. zero.) □

3. Uniform integrability and closure (case $\sum a_k^2 < \infty$).

Uniform integrability: a family bounded in L^2 is uniformly integrable. Indeed, by Markov then Cauchy-Schwarz, for all $K > 0$,

$$\mathbb{E}[|M_n| \mathbf{1}_{\{|M_n| > K\}}] \leq \frac{1}{K} \mathbb{E}[M_n^2 \mathbf{1}_{\{|M_n| > K\}}] \leq \frac{1}{K} \mathbb{E}[M_n^2] \leq \frac{C}{K},$$

where $C = \sup_n \mathbb{E}[M_n^2] < \infty$ by 2(a). The bound $C/K \rightarrow 0$ as $K \rightarrow \infty$, uniformly in n , whence the uniform integrability of (M_n) .

Closure: fix n . For $m > n$, the martingale property (by induction via the tower property) gives $\mathbb{E}[M_m | \mathcal{F}_n] = M_n$. Since $M_m \rightarrow M_\infty$ in L^1 (by 2(a)) and $X \mapsto \mathbb{E}[X | \mathcal{F}_n]$ is an L^1 contraction ($\|\mathbb{E}[X | \mathcal{F}_n]\|_1 \leq \|X\|_1$ by conditional Jensen), we have $\mathbb{E}[M_m | \mathcal{F}_n] \rightarrow \mathbb{E}[M_\infty | \mathcal{F}_n]$ in L^1 . As the left-hand side equals M_n for all $m > n$, we obtain

$$M_n = \mathbb{E}[M_\infty | \mathcal{F}_n] \quad (n \in \mathbb{N}),$$

that is, (M_n) is closed by M_∞ . (This is also Lévy's / the closed martingale theorem, valid as soon as (M_n) is uniformly integrable.) $\square$

4. Doob's maximal bound (case $a_k = a$).

(M_n) is a positive martingale, hence a positive submartingale. Doob's maximal inequality (Theorem 5.2(i)) gives, for all $\lambda > 0$,

$$\mathbb{P}\left(\max_{0 \leq k \leq n} M_k \geq \lambda\right) \leq \frac{1}{\lambda} \mathbb{E}[M_n \mathbf{1}_{\{\max_{0 \leq k \leq n} M_k \geq \lambda\}}] \leq \frac{1}{\lambda} \mathbb{E}[M_n] = \frac{1}{\lambda},$$

using $\mathbb{E}[M_n] = 1$ (question 1). This establishes the first inequality.

For the second, set $S = \sup_{k \geq 0} M_k$ and $S_n = \max_{0 \leq k \leq n} M_k$, an increasing sequence. Fix $\lambda > 0$. For all $\mu \in (0, \lambda)$, the events $\{S_n \geq \mu\}$ increase to $\bigcup_n \{S_n \geq \mu\} = \{S \geq \mu\}$ (if $S \geq \mu$ then some $M_k \geq \mu$, so $S_n \geq \mu$ as soon as $n \geq k$). By continuity from below of the measure and the first inequality,

$$\mathbb{P}(S \geq \mu) = \lim_{n \rightarrow \infty} \mathbb{P}(S_n \geq \mu) \leq \frac{1}{\mu}.$$

Since $\{S \geq \lambda\} \subset \{S \geq \mu\}$, we have $\mathbb{P}(S \geq \lambda) \leq 1/\mu$ for all $\mu < \lambda$; letting $\mu \uparrow \lambda$,

$$\mathbb{P}\left(\sup_{k \geq 0} M_k \geq \lambda\right) = \mathbb{P}(S \geq \lambda) \leq \frac{1}{\lambda}.$$

Consequence: $\mathbb{P}(S = \infty) \leq \mathbb{P}(S \geq \lambda) \leq 1/\lambda$ for all λ , so letting $\lambda \rightarrow \infty$, $\mathbb{P}(S = \infty) = 0$: $\sup_{k \geq 0} M_k < \infty$ almost surely. $\square$

5. The degenerate constant case $a_k = a \in (0, 1)$.

(a) Since $1 + a\xi_k \in \{1 - a, 1 + a\} \subset (0, 2)$, we may take the logarithm: $L_n = \log M_n = \sum_{k=1}^n Y_k$ with $Y_k = \log(1 + a\xi_k)$. The Y_k are i.i.d., bounded (values $\log(1 \pm a)$, finite), hence integrable, with mean

$$\mu := \mathbb{E}[Y_k] = \frac{1}{2} \log(1 + a) + \frac{1}{2} \log(1 - a) = \frac{1}{2} \log(1 - a^2).$$

By the provided inequality ($\frac{1}{2} \log(1 - x^2) \leq -\frac{x^2}{2}$, which follows from $\log(1 - u) \leq -u$ with $u = x^2$), we have $\mu \leq -a^2/2 < 0$ since $a > 0$. By the strong law of large numbers (the Y_k being i.i.d. and integrable),

$$\frac{L_n}{n} = \frac{1}{n} \sum_{k=1}^n Y_k \xrightarrow[n \rightarrow \infty]{\text{a.s.}} \mu < 0,$$

so $L_n \rightarrow -\infty$ a.s., that is, $M_n = e^{L_n} \rightarrow 0$ almost surely. Since $M_n \rightarrow M_\infty$ a.s. by 1, we conclude $M_\infty = 0$ a.s.

(b) We have $\mathbb{E}[M_n] = 1$ for all n (question 1) but $\mathbb{E}[M_\infty] = 0$, so $\mathbb{E}[M_n] \not\rightarrow \mathbb{E}[M_\infty]$. If (M_n) were uniformly integrable, it would converge in L^1 (Vitali's theorem) to its a.s. limit M_∞ , which would force $\mathbb{E}[M_n] \rightarrow \mathbb{E}[M_\infty]$, absurd. Hence (M_n) is *not uniformly integrable*, and it *does not converge in*

L^1 (otherwise the L^1 limit would coincide with the a.s. limit $M_\infty = 0$, again yielding $\mathbb{E}[M_n] \rightarrow 0$). Finally it is not closed: $\mathbb{E}[M_\infty \mid \mathcal{F}_n] = \mathbb{E}[0 \mid \mathcal{F}_n] = 0 \neq M_n$. This is therefore an example of a martingale bounded in L^1 , convergent a.s., but *not closed*. $\square$

6. General case $\sum_k a_k^2 = \infty$ (Kakutani).

Since $\sqrt{1 + a_k \xi_k} > 0$ is bounded and a function of ξ_k , set $c_k = \mathbb{E}[\sqrt{1 + a_k \xi_k}] = \frac{1}{2}(\sqrt{1 + a_k} + \sqrt{1 - a_k}) > 0$ and $N_n = \prod_{k=1}^n \sqrt{1 + a_k \xi_k} / c_k$. Then $N_n > 0$, $\mathcal{F}_n$ -measurable and bounded, and by the same computation as in 1 (independence, taking out what is measurable),

$$\mathbb{E}[N_n \mid \mathcal{F}_{n-1}] = N_{n-1} \frac{\mathbb{E}[\sqrt{1 + a_n \xi_n}]}{c_n} = N_{n-1},$$

so (N_n) is a positive martingale, $\mathbb{E}[N_n] = 1$. By 1 (applied to (N_n)), $N_n \rightarrow N_\infty$ a.s. with $N_\infty < \infty$ a.s.

We compute $c_k^2 = \frac{1}{4}(2 + 2\sqrt{1 - a_k^2}) = \frac{1}{2}(1 + \sqrt{1 - a_k^2})$, whence

$$1 - c_k^2 = \frac{1}{2}(1 - \sqrt{1 - a_k^2}) = \frac{1}{2} \cdot \frac{a_k^2}{1 + \sqrt{1 - a_k^2}} \geq \frac{a_k^2}{4},$$

since $1 + \sqrt{1 - a_k^2} \leq 2$. Hence $\sum_k (1 - c_k^2) \geq \frac{1}{4} \sum_k a_k^2 = \infty$. Since $c_k \in (0, 1]$ and $\log c_k^2 \leq c_k^2 - 1$, we have $\sum_k \log c_k^2 \leq -\sum_k (1 - c_k^2) = -\infty$, so $\prod_{k=1}^n c_k \rightarrow 0$.

But by construction $\sqrt{M_n} = N_n \prod_{k=1}^n c_k$. Since $N_n \rightarrow N_\infty < \infty$ a.s. and $\prod_{k=1}^n c_k \rightarrow 0$, we obtain $\sqrt{M_n} \rightarrow 0$ a.s., so $M_\infty = 0$ almost surely. (The consequences of 5(b) — non-uniform-integrability, non-convergence in L^1 , non-closure — remain valid identically as soon as $\sum a_k^2 = \infty$.) $\square$

Remark (Kakutani's dichotomy). We have the dichotomy: $\mathbb{E}[M_\infty] = 1$ and (M_n) closed if $\sum a_k^2 < \infty$; $M_\infty = 0$ a.s. and (M_n) not uniformly integrable if $\sum a_k^2 = \infty$. This is a particular case of Kakutani's theorem for product martingales of independent positive factors with mean 1.

Problem IV — Geometric Bienaymé–Galton–Watson process: explicit law of the martingale limit W (21 pts)

Question 1.

(a) Geometric distribution: $f(s) = \sum_{k \geq 0} (1-p)p^k s^k = (1-p) \sum_{k \geq 0} (ps)^k = \frac{1-p}{1-ps}$, the geometric series converging for $s \in [0, 1]$ since $ps < 1$.

The mean equals $m = f'(1)$. We have $f'(s) = \frac{(1-p)p}{(1-ps)^2}$, whence

$$m = f'(1) = \frac{(1-p)p}{(1-p)^2} = \frac{p}{1-p}.$$

Since $p > \frac{1}{2}$, we have $p > 1-p$ so $m > 1$ (*supercritical* process).

For the variance, $f''(s) = \frac{2(1-p)p^2}{(1-ps)^3}$, so $f''(1) = \frac{2(1-p)p^2}{(1-p)^3} = \frac{2p^2}{(1-p)^2} = 2m^2$. But $\mathbb{E}[\xi(\xi-1)] = f''(1)$, whence $\mathbb{E}[\xi^2] = 2m^2 + m$ and

$$\sigma^2 = \mathbb{E}[\xi^2] - m^2 = m^2 + m = \frac{p^2}{(1-p)^2} + \frac{p}{1-p} = \frac{p}{(1-p)^2}.$$

(We keep the compact form $\sigma^2 = m^2 + m = m(m+1)$.)

(b) Conditionally on $\mathcal{F}_n$, Z_n is $\mathcal{F}_n$ -measurable and $(\xi_i^{(n)})_i$ is independent of $\mathcal{F}_n$, so by the conditional Wald formula

$$\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] = \mathbb{E}\left[\sum_{i=1}^{Z_n} \xi_i^{(n)} \mid \mathcal{F}_n\right] = m Z_n.$$

Hence

$$\mathbb{E}[W_{n+1} \mid \mathcal{F}_n] = \frac{\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n]}{m^{n+1}} = \frac{m Z_n}{m^{n+1}} = \frac{Z_n}{m^n} = W_n.$$

Since $W_n \geq 0$ and $\mathbb{E}[W_n] = \mathbb{E}[W_0] = 1 < \infty$, (W_n) is a positive martingale, in particular a positive supermartingale bounded in L^1 ($\sup_n \mathbb{E}[W_n] = 1$). By the almost sure convergence theorem for (super)martingales (Thm 5.1, consequence of Doob's upcrossing inequality), $W_n \rightarrow W$ a.s. with $W \geq 0$, and by Fatou's lemma $\mathbb{E}[W] \leq \liminf_n \mathbb{E}[W_n] = 1$.

Question 2.

(a) The extinction probability q is the smallest fixed point of f in $[0, 1]$ (Thm 1.2: f is convex and increasing on $[0, 1]$ with $f(1) = 1$; in the supercritical regime $m = f'(1) > 1$, this smallest fixed point is < 1). Let us solve:

$$\frac{1-p}{1-ps} = s \iff 1-p = s(1-ps) = s - ps^2 \iff ps^2 - s + (1-p) = 0.$$

The discriminant equals $\Delta = 1 - 4p(1-p) = (1-2p)^2$, so $\sqrt{\Delta} = 2p-1$ (since $p > \frac{1}{2}$). The roots are

$$s = \frac{1 \pm (2p-1)}{2p} \in \left\{ 1, \frac{1-p}{p} \right\}.$$

The smallest is $q = \frac{1-p}{p}$. Since $m = \frac{p}{1-p}$, we indeed have $q = \frac{1}{m} \in (0, 1)$.

(b) Case $n = 1$. Let us show $\frac{f(s)-1}{f(s)-q} = m \frac{s-1}{s-q}$. We compute

$$f(s) - 1 = \frac{1-p}{1-ps} - 1 = \frac{(1-p) - (1-ps)}{1-ps} = \frac{p(s-1)}{1-ps},$$

$$f(s) - q = \frac{1-p}{1-ps} - \frac{1-p}{p} = \frac{p(1-p) - (1-p)(1-ps)}{p(1-ps)} = \frac{(1-p)(p-1+ps)}{p(1-ps)}.$$

But $p-1+ps = p(s - \frac{1-p}{p}) = p(s-q)$, whence $f(s) - q = \frac{(1-p)p(s-q)}{p(1-ps)} = \frac{(1-p)(s-q)}{1-ps}$. The ratio gives

$$\frac{f(s) - 1}{f(s) - q} = \frac{p(s-1)/(1-ps)}{(1-p)(s-q)/(1-ps)} = \frac{p}{1-p} \cdot \frac{s-1}{s-q} = m \frac{s-1}{s-q}.$$

Iteration. Set $g(s) = \frac{s-1}{s-q}$. The above equality reads $g(f(s)) = m g(s)$. Since $f_n = f \circ f_{n-1}$ and $f_n(s) \in [0, 1)$ for $s \in [0, 1)$ (because $f([0, 1)) \subset [0, 1)$), we obtain by induction

$$g(f_n(s)) = m g(f_{n-1}(s)) = \dots = m^n g(s),$$

that is, $\frac{f_n(s) - 1}{f_n(s) - q} = m^n \frac{s-1}{s-q}$.

Closed form. Setting $A := m^n \frac{s-1}{s-q}$ and solving $\frac{f_n - 1}{f_n - q} = A$:

$$f_n - 1 = A(f_n - q) \Rightarrow f_n(1 - A) = 1 - Aq \Rightarrow f_n(s) = \frac{1 - Aq}{1 - A} \quad (*)$$

With $q = 1/m$ and $A = m^n \frac{s-1}{s-1/m} = m^n \frac{m(s-1)}{ms-1}$, we multiply numerator and denominator by $(ms-1)$ in $(*)$:

$$f_n(s) = \frac{(ms-1) - m^n(s-1)}{(ms-1) - m^{n+1}(s-1)} = \frac{(m^n-1) - (m^n-m)s}{(m^{n+1}-1) - (m^{n+1}-m)s},$$

after grouping the constants and the terms in s . This is the announced formula. One checks $f_n(1) = 1$ and $f_n(0) = \frac{m^n-1}{m^{n+1}-1} \xrightarrow{n \rightarrow \infty} \frac{1}{m} = q$: this is $\mathbb{P}(Z_n = 0) \uparrow q$, consistent.

Question 3.

(a) For $\lambda \geq 0$, set $s = e^{-\lambda/m^n} \in (0, 1]$. Since $W_n = Z_n/m^n$, we have $e^{-\lambda W_n} = s^{Z_n}$, whence $\mathbb{E}[e^{-\lambda W_n}] = \mathbb{E}[s^{Z_n}] = f_n(s)$. Since $W_n \rightarrow W$ a.s. and $0 \leq e^{-\lambda W_n} \leq 1$, the dominated convergence theorem gives

$$\mathbb{E}[e^{-\lambda W}] = \lim_{n \rightarrow \infty} f_n(e^{-\lambda/m^n}).$$

Set $\varepsilon = m^{-n} \rightarrow 0$ and expand $s = e^{-\lambda\varepsilon} = 1 - \lambda\varepsilon + o(\varepsilon)$, whence $s-1 = -\lambda\varepsilon + o(\varepsilon)$. Then

$$A = m^n \frac{m(s-1)}{ms-1} = \frac{1}{\varepsilon} \cdot \frac{m(s-1)}{ms-1}.$$

But $ms-1 = m(1-\lambda\varepsilon)-1+o(\varepsilon) = (m-1)-m\lambda\varepsilon+o(\varepsilon) \rightarrow m-1$, and $\frac{m(s-1)}{\varepsilon} = \frac{m(-\lambda\varepsilon+o(\varepsilon))}{\varepsilon} \rightarrow -m\lambda$. Therefore

$$A \xrightarrow{n \rightarrow \infty} \frac{-m\lambda}{m-1}.$$

With $f_n = \frac{1-Aq}{1-A} = \frac{1-A/m}{1-A}$ and passing to the limit (the function $A \mapsto \frac{1-A/m}{1-A}$ being continuous at the limit point, which is $< 0 \neq 1$),

$$\mathbb{E}[e^{-\lambda W}] = \frac{1 - \frac{1}{m} \cdot \frac{-m\lambda}{m-1}}{1 - \frac{-m\lambda}{m-1}} = \frac{1 + \frac{\lambda}{m-1}}{1 + \frac{m\lambda}{m-1}} = \frac{(m-1) + \lambda}{(m-1) + m\lambda} = \frac{\lambda + m - 1}{m\lambda + m - 1}.$$

(b) Write $a = \frac{1}{m}$ and $b = \frac{m-1}{m}$. We verify the identity

$$\frac{\lambda + m - 1}{m\lambda + m - 1} = a + (1 - a) \frac{b}{b + \lambda}.$$

Indeed, $a + (1 - a) \frac{b}{b + \lambda} = \frac{1}{m} + \frac{m-1}{m} \cdot \frac{(m-1)/m}{(m-1)/m + \lambda} = \frac{1}{m} + \frac{m-1}{m} \cdot \frac{m-1}{(m-1) + m\lambda}$, and reducing to the common denominator $m((m-1) + m\lambda)$:

$$= \frac{(m-1) + m\lambda + (m-1)^2}{m((m-1) + m\lambda)} = \frac{m(m-1) + m\lambda}{m((m-1) + m\lambda)} = \frac{(m-1) + \lambda}{(m-1) + m\lambda},$$

which is indeed the transform found. Now $\lambda \mapsto \frac{b}{b + \lambda}$ is the Laplace transform of an $\mathcal{E}(b)$ distribution, and the constant 1 is that of the mass δ_0 . By uniqueness of the Laplace transform on distributions supported on $[0, \infty)$, W is the mixture

$$W \stackrel{d}{=} a \delta_0 + (1 - a) \mathcal{E}(b), \quad a = \frac{1}{m}, \quad b = \frac{m-1}{m}.$$

Hence $\mathbb{P}(W = 0) = a = \frac{1}{m} = q$, and conditionally on $\{W > 0\}$, $W \sim \mathcal{E}(\frac{m-1}{m})$. (The fact $\mathbb{P}(W = 0) = q$ illustrates that $\{W = 0\}$ coincides a.s. with extinction.)

Mean: with $\psi(\lambda) = \frac{\lambda + m - 1}{m\lambda + m - 1}$,

$$\psi'(\lambda) = \frac{(m\lambda + m - 1) - m(\lambda + m - 1)}{(m\lambda + m - 1)^2} = \frac{(m-1) - m(m-1)}{(m\lambda + m - 1)^2} = \frac{-(m-1)^2}{(m\lambda + m - 1)^2},$$

so $\psi'(0) = \frac{-(m-1)^2}{(m-1)^2} = -1$ and $\mathbb{E}[W] = -\psi'(0) = 1$. Since $\mathbb{E}[W] = 1 = \mathbb{E}[W_0]$, the martingale is not degenerate: $\mathbb{P}(W > 0) = 1 - q > 0$.

Question 4.

(a) Conditionally on $\mathcal{F}_n$, $Z_{n+1} = \sum_{i=1}^{Z_n} \xi_i^{(n)}$ is a sum of Z_n i.i.d. terms with mean m and variance σ^2 , independent of $\mathcal{F}_n$, and Z_n is $\mathcal{F}_n$ -measurable. Hence

$$\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] = m Z_n, \quad \text{Var}(Z_{n+1} \mid \mathcal{F}_n) = \sigma^2 Z_n,$$

whence $\mathbb{E}[Z_{n+1}^2 \mid \mathcal{F}_n] = \text{Var}(Z_{n+1} \mid \mathcal{F}_n) + (\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n])^2 = \sigma^2 Z_n + m^2 Z_n^2$. Dividing by $m^{2(n+1)}$ and taking the expectation (with $\mathbb{E}[Z_n] = m^n$):

$$\mathbb{E}[W_{n+1}^2] = \frac{\mathbb{E}[Z_{n+1}^2]}{m^{2n+2}} = \frac{m^2 \mathbb{E}[Z_n^2] + \sigma^2 \mathbb{E}[Z_n]}{m^{2n+2}} = \mathbb{E}[W_n^2] + \frac{\sigma^2 m^n}{m^{2n+2}} = \mathbb{E}[W_n^2] + \frac{\sigma^2}{m^{n+2}}.$$

(b) Summing from $W_0^2 = 1$ (since $Z_0 = 1$):

$$\mathbb{E}[W_n^2] = 1 + \sigma^2 \sum_{k=0}^{n-1} \frac{1}{m^{k+2}} = 1 + \frac{\sigma^2}{m^2} \sum_{k=0}^{n-1} m^{-k} \xrightarrow{n \rightarrow \infty} 1 + \frac{\sigma^2}{m^2} \cdot \frac{1}{1 - 1/m} = 1 + \frac{\sigma^2}{m(m-1)},$$

the geometric series converging since $m > 1$. Hence $\sup_n \mathbb{E}[W_n^2] = 1 + \frac{\sigma^2}{m(m-1)} < \infty$: the martingale (W_n) is bounded in L^2 . By the L^p convergence theorem for martingales ($p = 2$, Thm 5.3, via Doob's maximal inequality), $W_n \rightarrow W$ in L^2 (and hence in L^1 , which reconfirms $\mathbb{E}[W] = 1$).

The L^2 convergence gives $\mathbb{E}[W^2] = \lim_n \mathbb{E}[W_n^2] = 1 + \frac{\sigma^2}{m(m-1)}$, whence, with $\mathbb{E}[W] = 1$,

$$\text{Var}(W) = \mathbb{E}[W^2] - 1 = \frac{\sigma^2}{m(m-1)}.$$

With $\sigma^2 = m^2 + m = m(m+1)$ (Question 1a),

$$\boxed{\text{Var}(W) = \frac{m(m+1)}{m(m-1)} = \frac{m+1}{m-1}} \quad \left(= \frac{1}{2p-1} \right).$$

Consistency with Question 3. For the mixture $W \stackrel{d}{=} a\delta_0 + (1-a)\mathcal{E}(b)$ with $a = 1/m$, $b = (m-1)/m$: since $\mathbb{E}[\mathcal{E}(b)^2] = 2/b^2$,

$$\mathbb{E}[W^2] = (1-a) \cdot \frac{2}{b^2} = \left(1 - \frac{1}{m}\right) \cdot \frac{2m^2}{(m-1)^2} = \frac{m-1}{m} \cdot \frac{2m^2}{(m-1)^2} = \frac{2m}{m-1}.$$

Hence $\text{Var}(W) = \frac{2m}{m-1} - 1 = \frac{2m-(m-1)}{m-1} = \frac{m+1}{m-1}$, identical to the value above. The two approaches agree.

Remark. We obtain a fully explicit example where the limit of the fundamental martingale of a supercritical branching process has a known law (a mixture of a point mass at 0 – extinction – and an exponential), with $\mathbb{P}(W=0) = q$ and $\mathbb{E}[W] = 1$. The L^2 bound established in Q4 directly provides non-degeneracy in this setting (where ξ has moments of all orders).

Problem V — Brownian Motion with Negative Drift: Crossing Probability and the Law of the Maximum (20 pts)

Notation and reminders. We write $M_t = M_t^\lambda$ when λ is fixed. We freely use the facts from the course: for $0 \leq s \leq t$, the increment $B_t - B_s$ is independent of $\mathcal{F}_s$ and has distribution $\mathcal{N}(0, t - s)$; the Gaussian Laplace transform $\mathbb{E}[e^{\mu Z}] = e^{\mu^2 \sigma^2 / 2}$ for $Z \sim \mathcal{N}(0, \sigma^2)$; the strong law of large numbers for Brownian motion $B_t/t \rightarrow 0$ a.s. ($t \rightarrow \infty$); the optional stopping theorem at a bounded stopping time and the strong Markov property.

Question 1.

(a) Fix $\lambda \in \mathbb{R}$. For all t , $B_t \sim \mathcal{N}(0, t)$ hence

$$\mathbb{E}[M_t^\lambda] = e^{-\lambda^2 t / 2} \mathbb{E}[e^{\lambda B_t}] = e^{-\lambda^2 t / 2} e^{\lambda^2 t / 2} = 1 < \infty,$$

which proves integrability and the last assertion. The process (M_t^λ) is adapted since B_t is. Let $0 \leq s \leq t$. Write $B_t = B_s + (B_t - B_s)$; since $B_t - B_s$ is independent of $\mathcal{F}_s$ and $\sim \mathcal{N}(0, t - s)$, and $e^{\lambda B_s}$ is $\mathcal{F}_s$ -measurable,

$$\mathbb{E}[M_t^\lambda \mid \mathcal{F}_s] = e^{-\lambda^2 t / 2} e^{\lambda B_s} \mathbb{E}[e^{\lambda(B_t - B_s)} \mid \mathcal{F}_s] = e^{-\lambda^2 t / 2} e^{\lambda B_s} e^{\lambda^2(t-s)/2} = e^{\lambda B_s - \lambda^2 s / 2} = M_s^\lambda,$$

where we used independence to replace the conditional expectation of $e^{\lambda(B_t - B_s)}$ by its expectation. Hence (M_t^λ) is a martingale.

(b) By assumption $B_t/t \rightarrow 0$ a.s. as $t \rightarrow \infty$. Hence, almost surely,

$$X_t = B_t - ct = t \left(\frac{B_t}{t} - c \right) \xrightarrow[t \rightarrow \infty]{} -\infty,$$

since $c > 0$ implies $\frac{B_t}{t} - c \rightarrow -c < 0$. On the probability-1 event where this limit equals $-\infty$, the function $t \mapsto X_t$ is continuous on $[0, \infty)$ and tends to $-\infty$; it is therefore bounded above (it even attains its maximum), so that $S = \sup_{t \geq 0} X_t < \infty$. Finally $S \geq X_0 = B_0 - 0 = 0$, whence $S \geq 0$. $\square$

Question 2.

(a) The process (X_t) is continuous and adapted; $\tau_a = \inf\{t : X_t \geq a\}$ is the first hitting time of the closed set $[a, \infty)$ by a continuous adapted process, hence a stopping time (with the convention $\inf \emptyset = +\infty$). For each fixed t , $\tau_a \wedge t$ is a *bounded* stopping time (bounded by t). Take $\lambda = 2c > 0$; the martingale is then

$$M_t^{2c} = \exp(2c B_t - 2c^2 t) = \exp(2c(B_t - ct)) = \exp(2c X_t).$$

By the optional stopping theorem applied to the martingale (M_t^{2c}) and the bounded stopping time $\tau_a \wedge t$,

$$\mathbb{E}[M_{\tau_a \wedge t}^{2c}] = \mathbb{E}[M_0^{2c}] = \mathbb{E}[1] = 1,$$

that is, $\mathbb{E}[\exp(2c B_{\tau_a \wedge t} - 2c^2(\tau_a \wedge t))] = 1$, which is the desired identity (the integrand equals $e^{2c X_{\tau_a \wedge t}}$).

(b) Set $Y_t := M_{\tau_a \wedge t}^{2c} = e^{2c X_{\tau_a \wedge t}}$.

Domination. By definition of τ_a as the *first* time X reaches a , and by continuity of X starting from $X_0 = 0 < a$, we have $X_s \leq a$ for all $s \leq \tau_a$ (and $X_{\tau_a} = a$ if $\tau_a < \infty$). Since $\tau_a \wedge t \leq \tau_a$, it follows that $X_{\tau_a \wedge t} \leq a$, hence

$$0 \leq Y_t \leq e^{2ca} \quad \text{for all } t,$$

which provides a domination by the integrable constant e^{2ca} .

Almost sure limit. We study Y_t as $t \rightarrow \infty$:

- On $\{\tau_a < \infty\}$: for $t \geq \tau_a$ we have $\tau_a \wedge t = \tau_a$, hence $Y_t = e^{2cX_{\tau_a}} = e^{2ca}$ (since $X_{\tau_a} = a$ by continuity). Thus $Y_t \rightarrow e^{2ca}$.
- On $\{\tau_a = \infty\}$: then $\tau_a \wedge t = t$ and $Y_t = e^{2cX_t}$. But, by Question 1(b), $X_t \rightarrow -\infty$ a.s., hence $Y_t = e^{2cX_t} \rightarrow 0$.

We thus have, almost surely, $Y_t \xrightarrow[t \rightarrow \infty]{} e^{2ca} \mathbf{1}_{\{\tau_a < \infty\}}$.

Since $0 \leq Y_t \leq e^{2ca}$, the dominated convergence theorem gives

$$1 = \lim_{t \rightarrow \infty} \mathbb{E}[Y_t] = \mathbb{E}[e^{2ca} \mathbf{1}_{\{\tau_a < \infty\}}] = e^{2ca} \mathbb{P}(\tau_a < \infty).$$

Hence

$$\boxed{\mathbb{P}(\tau_a < \infty) = e^{-2ca}.}$$

□

Question 3.

(a) We show $\{S \geq a\} = \{\tau_a < \infty\}$.

- ($\supseteq$) If $\tau_a < \infty$, then $X_{\tau_a} = a$, hence $S = \sup_t X_t \geq a$: thus $\{\tau_a < \infty\} \subseteq \{S \geq a\}$.
- ($\subseteq$) Suppose $S \geq a$. By Question 1(b), a.s. X is continuous on $[0, \infty)$ and $X_t \rightarrow -\infty$; such a function attains its maximum at a finite point t^* , that is, $X_{t^*} = S \geq a$. Since $X_0 = 0 < a \leq X_{t^*}$ and X is continuous, the intermediate value theorem provides an instant $u \in [0, t^*]$ with $X_u = a$; hence $\tau_a \leq u < \infty$. Thus $\{S \geq a\} \subseteq \{\tau_a < \infty\}$.

This reasoning holds for any level $x > 0$ (replacing a by x , Question 2 being identical), hence

$$\mathbb{P}(S \geq x) = \mathbb{P}(\tau_x < \infty) = e^{-2cx}, \quad x > 0.$$

Since moreover $\mathbb{P}(S \geq 0) = 1$ (Question 1(b)), the survival function of S is $\mathbb{P}(S \geq x) = e^{-2cx}$ for all $x \geq 0$: this is exactly the tail of an exponential distribution. Therefore

$$\boxed{S \sim \mathcal{E}(2c),}$$

an exponential distribution with parameter $2c$ (density $2c e^{-2cx} \mathbf{1}_{x>0}$).

(b) For an $\mathcal{E}(2c)$ distribution,

$$\mathbb{E}[S] = \frac{1}{2c}, \quad \mathbb{P}(S = 0) = 1 - \lim_{x \downarrow 0} \mathbb{P}(S \geq x) = 1 - \lim_{x \downarrow 0} e^{-2cx} = 1 - 1 = 0.$$

The exponential distribution being atomless, $S > 0$ almost surely. □

Question 4.

Recall the identity $\{S \geq b\} = \{\tau_b < \infty\}$ (Question 3(a)) and $\mathbb{P}(\tau_b < \infty) = e^{-2cb}$ (admitted).

Idea. Conditionally on $\{S \geq a\} = \{\tau_a < \infty\}$, we "restart" the process at the instant τ_a , where $X_{\tau_a} = a$, and we ask whether it rises further by a height b .

On $\{\tau_a < \infty\}$, set for $t \geq 0$

$$\tilde{B}_t := B_{\tau_a+t} - B_{\tau_a}.$$

By the strong Markov property of Brownian motion applied to the stopping time τ_a , the process $(\tilde{B}_t)_{t \geq 0}$ is a standard Brownian motion, *independent* of $\mathcal{F}_{\tau_a}$. For $t \geq 0$, since $X_{\tau_a} = a$,

$$X_{\tau_a+t} = B_{\tau_a+t} - c(\tau_a + t) = (B_{\tau_a} - c\tau_a) + (\tilde{B}_t - ct) = a + \tilde{X}_t,$$

where $\tilde{X}_t := \tilde{B}_t - ct$ is a Brownian motion with drift $-c$ starting from 0, hence with the *same distribution* as X , and *independent* of $\mathcal{F}_{\tau_a}$.

Set $\tilde{S} := \sup_{t \geq 0} \tilde{X}_t$. Since $X_s \leq a$ for all $s \leq \tau_a$ (Question 2(b)), the global supremum decomposes, on $\{\tau_a < \infty\}$,

$$S = \sup_{t \geq 0} X_t = \max \left(\underbrace{\sup_{s \leq \tau_a} X_s}_{=a}, \sup_{t \geq 0} X_{\tau_a+t} \right) = \max(a, a + \tilde{S}) = a + \tilde{S},$$

the last equality coming from $\tilde{S} \geq \tilde{X}_0 = 0$. Therefore, on $\{\tau_a < \infty\} = \{S \geq a\}$,

$$\{S \geq a + b\} = \{\tilde{S} \geq b\}.$$

Now, by the strong Markov property, $\tilde{S}$ (which is a function of $(\tilde{X}_t)_{t \geq 0}$) is independent of $\mathcal{F}_{\tau_a}$ and has the same distribution as S , while the event $\{\tau_a < \infty\} = \{S \geq a\}$ belongs to $\mathcal{F}_{\tau_a}$. Hence

$$\mathbb{P}(S \geq a + b \mid S \geq a) = \frac{\mathbb{P}(\{\tau_a < \infty\} \cap \{\tilde{S} \geq b\})}{\mathbb{P}(\tau_a < \infty)} = \frac{\mathbb{P}(\tau_a < \infty) \mathbb{P}(\tilde{S} \geq b)}{\mathbb{P}(\tau_a < \infty)} = \mathbb{P}(\tilde{S} \geq b).$$

Since $\tilde{S} \stackrel{d}{=} S$, we conclude

$$\boxed{\mathbb{P}(S \geq a + b \mid S \geq a) = \mathbb{P}(S \geq b).}$$

Consistency with Question 3. The exponential distribution is memoryless:

$$\mathbb{P}(S \geq a + b \mid S \geq a) = \frac{e^{-2c(a+b)}}{e^{-2ca}} = e^{-2cb} = \mathbb{P}(S \geq b),$$

which indeed coincides with the identity above. Conversely, the strong Markov argument provides a *direct* probabilistic proof of the memorylessness of S , hence a structural reason why the law of the maximum of a Brownian motion with drift is exponential. $\square$